

MAYUR ATHAVALE

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SUMMARY

Software Engineer specializing in production AI systems and cloud-native platform engineering. Architected a **multi-workflow AI copilot** on LangGraph — 6-intent supervisor routing, hybrid RAG, HITL, think + write todos planning loop — lifting task accuracy to **98%+** and driving **200%+ lead growth** at Zotok.ai, self-hosted on **AWS ECS** with Terraform. Strong across TypeScript, Python, Go, and AWS; shipped multi-tenant GraphQL APIs, offline-first systems, and Terraform-managed infrastructure.

EXPERIENCE

- Software Development Engineer** **Jan 2025 – Present**
Hyderabad, India
Zotok.ai
 - Architected and shipped an **AI copilot** that attracted **100+ ICP customers**, increased daily sales leads by **200%+**, and drove enterprise demo bookings.
 - Designed a **think + write todos planning loop** — agent reasons via think tool, writes structured todos (pending/in_progress/completed) to state before executing tools; combined with **memory auto-compaction** (compress_node) this lifted task accuracy from **60–70% to 98%+**.
 - Built end-to-end **RAG pipeline**: designed OpenSearch indices, wrote **data ingestion workflows** for vector embeddings (1024-dim via AWS Bedrock), implemented **hybrid retrieval (KNN + BM25)**, and configured **Redis + RDS** as dual-layer checkpointer storage for fast session and persistent thread state.
 - Architected a **multi-workflow LangGraph supervisor** classifying queries into 6 intent types (search, payment, quotation, payment_collection, send_message, off_topic) with routing to specialized per-workflow agents; implemented **step-based HITL** 3-node task workflows (find – prepare – send) with interrupt/resume, **LangSmith** tracing, and **BigTool semantic registry** for dynamic tool scoping.
 - Self-hosted LangGraph on **AWS ECS** with FastAPI endpoints, **S3 + CloudFront CDN** for chatbot serving, and **Terraform-managed infrastructure** across ECS, RDS, SQS, SNS, DynamoDB, Amplify, and Secrets Manager.
 - Designed a **multi-tenant GraphQL layer** using **Hasura + NestJS** proxy with dynamic schema generation, query guards, and business logic middleware; built WhatsApp message flows via **SQS/SNS, DynamoDB Streams, Redington webhooks, and WAHA server**.
 - Led full-stack development of the **Typebot Builder** — designed complex multi-group connection UI with distinctive edge representations; built **Next.js 14 custom auth middleware**, user/workspace creation with **Prisma + MySQL RDS**, and **tRPC** handlers with complex Zod validation.
 - Built an **offline-first architecture** with Service Workers, custom caching, and background sync — reduced bundle from **100+ MB to 15 MB**; introduced **lazy chunking with Module Federation** cutting cold-start latency by **60%**; implemented a **custom priority queue** for offline-to-online sync handling **200+ queued API calls**.
 - Implemented **4 streaming strategies** (SSE, AG-UI protocol, AssistantUI, LangGraph multi-mode); built **React – AG-UI – FastAPI** streaming pipeline and **model-switching layer** across Azure GPT-4o, Gemini Flash, and Claude; owned end-to-end CI/CD cutting release cycles by **35%**.

PROJECTS & WRITING

- PR Manager** (Go, GitHub APIs, Actions) — CLI tool automating PR workflows: review assignment, merge queuing, and cross-platform binary releases.
- Terminal Portfolio** (Go, BubbleTea, Wish, Lipgloss, AWS EC2) — Interactive SSH-accessible TUI portfolio with multi-tab navigation, YAML-driven content, visitor analytics via SQLite, and live presence tracking. Deployed on **AWS EC2** with **GitHub Actions CI/CD** auto-building and hot-deploying via SCP and systemd.
- Technical Writing** — Published articles on system security, cloud infrastructure, and networking on Medium, including self-hosted VPN tunneling on AWS.

TECHNICAL SKILLS

Languages: TypeScript, Python, Go, Rust, JavaScript, SQL, Bash

AI/LLM: LangGraph, LangSmith, AWS Bedrock, RAG Pipelines, Vector Embeddings, OpenSearch (KNN + BM25), Human-in-the-Loop (HITL), ReAct Agents, Multi-Workflow Agent Orchestration, AssistantUI, AG-UI, Prompt Engineering, Multi-Agent Systems

Backend: Node.js, NestJS, FastAPI, Hasura, GraphQL, tRPC, TypeORM, Prisma, Zod, REST APIs, Streaming APIs

Infrastructure: AWS (ECS, EC2, S3, CloudFront, Amplify, RDS, SQS, SNS, DynamoDB Streams, EventBridge, Lambda, Secrets Manager, Bedrock), Terraform, Infrastructure as Code (IaC), Docker, Container Orchestration, Caddy, CI/CD

Frontend: React, Next.js 14 (SSR/RSC/Page Router), Service Workers, Module Federation, Redux Toolkit, Tailwind

Databases: PostgreSQL, MySQL, Redis, DynamoDB, MongoDB, OpenSearch, Firebase

Concepts: System Design (LLD/HLD), SOLID, Multi-Tenant Architecture, Event-driven Architecture, Offline-first Systems, Hybrid Search, Observability, Distributed Systems

EDUCATION

- College of Engineering Pune (COEP)** **2020 – 2024**
B.Tech